CS228 : Logic for Computer Science

Module 3: Predicate Logic

Lecture 15: Soundness, Completeness, and Structures in FOL

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Topics Covered till date

Module 1: Propositional Logic

1. Syntax
2. Semantics
3. Formal proof systems
4. Normal Forms

Module 2: The Problem of Satisfiability

1. Propositional Resolution
2. DPLL
3. CDCL
4. SAT/SMT tool demo

Module 3: Predicate Logic

1. Syntax of predicate logic (Lecture 12)
2. Semantics of predicate logic (Lecture 13)
3. Natural deduction for FOL (Lecture 14)
4. Today: is our proof system sound and complete, and just how powerful is FOL?

Quick Recap: Natural Deduction for FOL

- All proof rules for Propositional Logic carry over and remain sound for FOL.

$$\frac{}{t = t}\ =_i \qquad \frac{t_1 = t_2 \qquad \varphi[t_1/x]}{\varphi[t_2/x]}\ =_e$$

Equality: reflexivity axiom, and substitute equals for equals (terms only, and only where free for x).

$$\frac{\forall x \varphi}{\varphi[t/x]}\ \forall x\ e \qquad \frac{\begin{array}{|l} x_0 \\ \vdots \\ \varphi[x_0/x] \end{array}}{\forall x \varphi}\ \forall x\ i$$

Universal quantifier: instantiate to any term t (free for x); or prove $\varphi[x_0/x]$ for a fresh x_0 inside a box.

$$\frac{\varphi[t/x]}{\exists x \varphi}\ \exists x\ i \qquad \frac{\exists x \varphi \qquad \begin{array}{|ll} x_0 & \varphi[x_0/x] \\ & \vdots \\ & \chi \end{array}}{\chi}\ \exists x\ e$$

Existential quantifier: generalize $\varphi[t/x]$ to $\exists x \varphi$; or from $\exists x \varphi$ and a boxed proof of χ from fresh $x_0, \varphi[x_0/x]$, conclude χ (provided χ doesn't mention x_0).

Side conditions on freshness/scope are important – violate them and rules become unsound!

Quick Recap: (Ungraded) Pop quiz!

Recall the exercise from last lecture: Prove the following sequents using ND for FOL:

1. $\forall x \varphi \vdash \exists x \varphi$
2. $\forall x(P(x) \to Q(x)), \exists x P(x) \vdash \exists x Q(x)$
3. $\exists x P(x), \forall x \forall y (P(x) \to Q(y)) \vdash \forall y Q(y)$
   Example: A factory's policy is "if any laptop in a batch has a faulty battery, recall every laptop in the batch." We're told some laptop has a faulty battery – so every laptop gets recalled!
   ($P(x)$: x has a faulty battery; $Q(y)$: y gets recalled.)
4. "Every prime number greater than 2 is odd. There exists a prime number greater than 2. So, there exists an odd number." First formalize this in FOL, then prove it using ND!

Which rules would you use first? When do you need a fresh variable and when do you need a box?

First Question: $\forall x \varphi \vdash \exists x \varphi$

1. $\forall x \varphi \vdash \exists x \varphi$

1.	$\forall x \varphi$	premise
2.	φ	$\forall x\ e\ 1$
3.	$\exists x \varphi$	$\exists x\ i\ 2$

(choose $t = x$ for both steps: $\varphi[x/x] = \varphi$, so this is legal)

Second Question: $\forall x(P(x) \to Q(x)), \exists x P(x) \vdash \exists x Q(x)$

1.	$\forall x(P(x) \to Q(x))$	premise
2.	$\exists x P(x)$	premise
3.	x_0 $P(x_0)$	assumption
4.	$P(x_0) \to Q(x_0)$	$\forall x$ *e* 1
5.	$Q(x_0)$	MP 4,3
6.	$\exists x Q(x)$	$\exists x$ *i* 5
7.	$\exists x Q(x)$	$\exists x$ *e* 2, 3-6

Wrong!

1.	$\forall x(P(x) \to Q(x))$	premise
2.	$\exists x P(x)$	premise
3.	x_0 $P(x_0)$	assumption
4.	$P(x_0) \to Q(x_0)$	$\forall x$ *e* 1
5.	$Q(x_0)$	MP 4,3
6.	$Q(x_0)$	$\exists x$ *e* 2, 3-5
7.	$\exists x Q(x)$	$\exists x$ *i* 6

The box on the right closes one line too early: line 6 ($Q(x_0)$) lets the fresh x_0 escape its own box before $\exists x$ *i* is applied outside – but χ in $\exists x$ *e* must not mention x_0!

Third Question: $\exists x P(x), \forall x \forall y (P(x) \to Q(y)) \vdash \forall y Q(y)$

3. $\exists x P(x), \forall x \forall y (P(x) \to Q(y)) \vdash \forall y Q(y)$

1.	$\exists x P(x)$	premise
2.	$\forall x \forall y (P(x) \to Q(y))$	premise
3.	y_0	
4.	x_0 $P(x_0)$	assumption
5.	$\forall y (P(x_0) \to Q(y))$	$\forall x$ e 2
6.	$P(x_0) \to Q(y_0)$	$\forall y$ e 5
7.	$Q(y_0)$	MP 6,4
8.	$Q(y_0)$	$\exists x$ e 1, 4-7
9.	$\forall y Q(y)$	$\forall y$ i 3-8

(outer box scoped by fresh y_0; inner box scoped by fresh x_0, assuming $P(x_0)$)

Fourth Question: Formalization & Proof

Every prime number greater than 2 is odd. There exists a prime number greater than 2. So, there exists an odd number.

Formalization

Let $P(x)$: "x is a prime number greater than 2"; $Q(x)$: "x is odd."

$\forall x(P(x) \to Q(x)), \exists x P(x) \vdash \exists x Q(x)$

1.	$\forall x(P(x) \to Q(x))$	premise
2.	$\exists x P(x)$	premise
3.	x_0 $P(x_0)$	assumption
4.	$P(x_0) \to Q(x_0)$	$\forall x$ *e* 1
5.	$Q(x_0)$	MP 4,3
6.	$\exists x Q(x)$	$\exists x$ *i* 5
7.	$\exists x Q(x)$	$\exists x$ *e* 2, 3-6

Topic 15.1

Soundness, Completeness, and the Power of FOL

Soundness and completeness

Soundness: If $\varphi \vdash \psi$, then $\varphi \models \psi$

- We can only derive semantically correct proofs!
- Intuitively easy to see, again by structural induction.

Is this proof system also complete?

Completeness: If $\varphi \models \psi$, then $\varphi \vdash \psi$

- Is anything that is true, derivable?
- Yes, but we won't stop to prove it here!

So even checking whether $\vdash \varphi$ holds is undecidable – because validity itself ($\models \varphi$) is undecidable, and completeness tells us $\vdash$ and $\models$ coincide!

How powerful is First-order logic?

Even basic properties are undecidable.

Can it express everything of interest?

How can we use it if it is so powerful?

Food for thought!

Topic 15.2

Structures and Expressiveness of FOL

Quick Recap: Semantics of Predicate Logic (Lecture 13)

- Vocabulary $\mathcal{V}$: the constants, functions, and predicates available (e.g. $\mathcal{V} = \{a, f, =, P\}$).
- $\mathcal{V}$-structure (or Model) $M = (U^M, (a^M, f^M, P^M))$: a universe U^M plus an interpretation of every symbol in $\mathcal{V}$ over it (e.g. $M = (\mathbb{N}, (0, +, <))$).
- Environment α: maps free variables to elements of U^M (needed only when φ has free variables).
- Satisfaction $M, \alpha \models \varphi$: defined by structural induction on φ.

First-order Definable Structures

- Recall that, if φ is a sentence (no free variables), no environment is necessary for evaluating truth of φ
  - So given $\mathcal{V}$-structure M, we can ask if $M \models \varphi$
  - Class of $\mathcal{V}$-structures defined by φ is just $\{M \mid M \models \varphi\}$
  - Such classes of structures that are defined by an FOL sentence are called FO-definable.
- We will see three commonly occurring examples of structures:
  - Numbers (we saw this already!)
  - Graphs
  - Databases

Number systems as FO structures

Natural/reals with add, mult, lin order and constants 0,1

- $\mathfrak{N} = \{(\mathbb{N}, 0, 1, \times, +, <, =)\}$
- $\mathfrak{R} = \{(\mathbb{R}, 0, 1, \times, +, <, =)\}$

Examples of properties expressible in FOL

- Every number has a cube root
- $\mathfrak{R} \models \forall x \exists y (x = ((y \times y) \times y))$
- $\mathfrak{N} \not\models \forall x \exists y (x = ((y \times y) \times y))$
- There are infinitely many prime natural numbers. **Exercise 15.1**

Maybe we can check validity, but given the model/structure can we check if it satisfies a given formula? Model checking!